Embedded OS Implementation, Fall 2026

Project #1 (due at November 5th, 2026 (Thursday) 08:00)

[PART I] Task Control Block Linked List

Objective:

Following the previous homework (HW1), please add some code to the µC/OS-II scheduler **in the kernel level** to observe the operations of the task control block (TCB) and TCB linked list.

※ The TCB address is dynamic.

※ Show work by attaching printing information and code submission, and **don't** need to be written to the output file.

The example output results are shown below:

```
OSTick    created, Thread ID 11760
Task[ 63] created, TCB Address  92e9e0
------After TCB[63] begin linked------
Previous TCB point to adress         0
Current  TCB point to adress    92e9e0
Next     TCB point to adress         0

The file 'TaskSet.txt' was opened
Task[  1] created, TCB Address  92ea54
------After TCB[ 1] begin linked------
Previous TCB point to adress         0
Current  TCB point to adress    92ea54
Next     TCB point to adress    92e9e0

Task[  2] created, TCB Address  92eac8
------After TCB[ 2] begin linked------
Previous TCB point to adress         0
Current  TCB point to adress    92eac8
Next     TCB point to adress    92ea54

================TCB linked list================
Task    Prev_TCB_addr   TCB_addr   Next_TCB_addr
 2                0       92eac8        92ea54
 1           92eac8       92ea54        92e9e0
63           92ea54       92e9e0             0
```

[PART II] RM Scheduler Implementation

Objective:

To implement the Rate Monotonic (RM) scheduler for periodic tasks and observe the scheduling behaviors.

Problem Definition:

Implement the following three task sets of periodic tasks. Add necessary code to the μC/OS-II scheduler **in the kernel level** to observe how the task **suffers** from the scheduler. We give the files for the parameter of the task.

Periodic Task Set = {τ_{ID} (ID, arrival time, execution time, period)}

Example Task Set 1 = {τ_1 (1, 0, 1, 3), τ_2 (2, 0, 3, 5)}

Example Task Set 2 = {τ_1 (1, 0, 3, 8), τ_2 (2, 1, 2, 6), τ_3 (3, 0, 4, 15)}:

Example Task Set 3 = {τ_1 (1, 1, 1, 6), τ_2 (2, 1, 1, 8), τ_3 (3, 5, 1, 12) , τ_4 (4, 0, 3, 12)}

※ If tasks have same period, the task with the smaller TaskID will be executed first.

※ The priority of the task is set according to the RM scheduling rules.

The input file format:

Task ID	Arrive Time	Execution Time	Task Period
##	##	##	##

Example of task set file 1:

```
1 0 1 3
2 0 3 5
```

Evaluation:

The output format:

Tick	Event	CurrentTask ID	NextTask ID	Response Time	Preemption Time	Preemption Instance
##	Preemption	task(ID)(job number)	task(ID)(job number)			
##	Completion	task(ID)(job number)	task(ID)(job number)	##	##	##
##	MissDeadline	task(ID)(job number)	----------			
##	task(ID) is running					

※ If the task is Idle Task, print "*task(priority)*".

Response Time: the duration between the task's arrival time and the time it is completed.

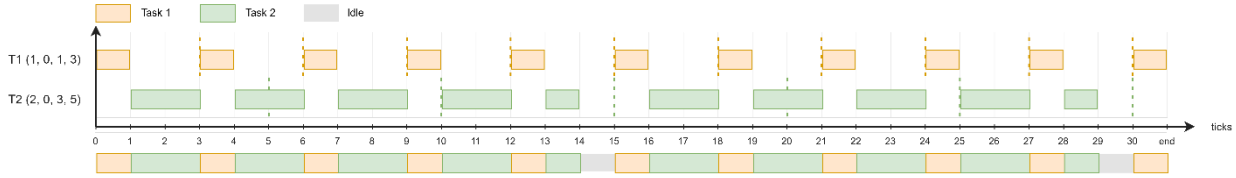
Preemption Time: the time this task is preempted by higher-priority tasks.

Preemption Instance: amount of preemption happened in this task

※ You have to print and output "task(ID) is running" message in task() function.

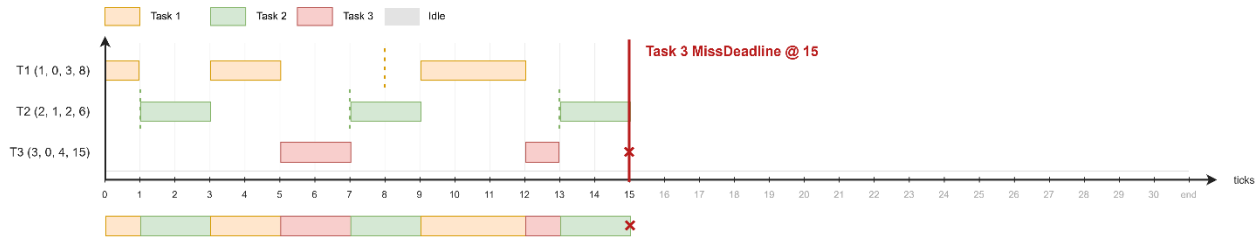
※ Output priority: MissDeadline > Preemption > Completion > task(ID) is running .

The output results of Task Set 1 = {τ_1 (1, 0, 1, 3), τ_2 (2, 0, 3, 5)}:



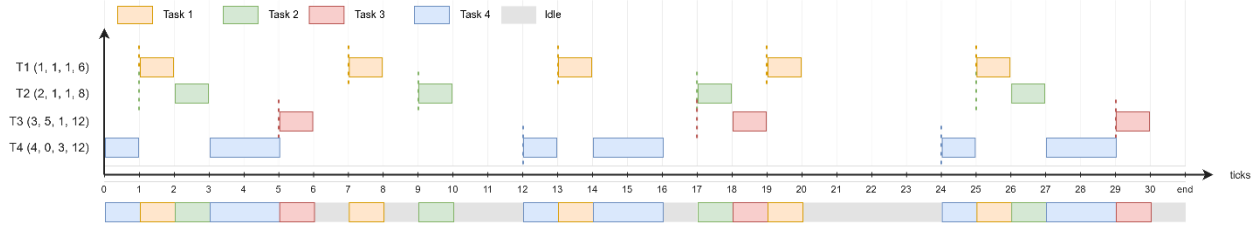

```
0   task( 1) is running
1   Completion  task( 1)( 0)    task( 2)( 0)    1   0   0
1   task( 2) is running
2   task( 2) is running
3   Preemption  task( 2)( 0)    task( 1)( 1)
3   task( 1) is running
4   Completion  task( 1)( 1)    task( 2)( 0)    1   0   0
4   task( 2) is running
5   Completion  task( 2)( 0)    task( 2)( 1)    5   1   1
5   task( 2) is running
6   Preemption  task( 2)( 1)    task( 1)( 2)
6   task( 1) is running
7   Completion  task( 1)( 2)    task( 2)( 1)    1   0   0
7   task( 2) is running
8   task( 2) is running
9   Completion  task( 2)( 1)    task( 1)( 3)    4   1   1
9   task( 1) is running
10  Completion  task( 1)( 3)    task( 2)( 2)    1   0   0
10  task( 2) is running
11  task( 2) is running
12  Preemption  task( 2)( 2)    task( 1)( 4)
12  task( 1) is running
13  Completion  task( 1)( 4)    task( 2)( 2)    1   0   0
13  task( 2) is running
14  Completion  task( 2)( 2)    task(63)    4   1   1
15  Preemption  task(63)    task( 1)( 5)
15  task( 1) is running
```

The output results of Task Set 2 = {τ_1 (1, 0, 3, 8), τ_2 (2, 1, 2, 6), τ_3 (3, 0, 4, 15)}:




```
0   task( 1) is running
1   Preemption  task( 1)( 0)    task( 2)( 0)
1   task( 2) is running
2   task( 2) is running
3   Completion  task( 2)( 0)    task( 1)( 0)    2   0   0
3   task( 1) is running
4   task( 1) is running
5   Completion  task( 1)( 0)    task( 3)( 0)    5   2   1
5   task( 3) is running
6   task( 3) is running
7   Preemption  task( 3)( 0)    task( 2)( 1)
7   task( 2) is running
8   task( 2) is running
9   Completion  task( 2)( 1)    task( 1)( 1)    2   0   0
9   task( 1) is running
10  task( 1) is running
11  task( 1) is running
12  Completion  task( 1)( 1)    task( 3)( 0)    4   0   0
12  task( 3) is running
13  Preemption  task( 3)( 0)    task( 2)( 2)
13  task( 2) is running
14  task( 2) is running
15  MissDeadline    task( 3)( 0)    ------------------
```

The output results of Task Set 3 = {τ_1 (1, 1, 1, 6), τ_2 (2, 1, 1, 8), τ_3 (3, 5, 1, 12) , τ_4 (4, 0, 3, 12)}:




```
0   task( 4) is running
1   Preemption  task( 4)( 0)    task( 1)( 0)
1   task( 1) is running
2   Completion  task( 1)( 0)    task( 2)( 0)    1   0   0
2   task( 2) is running
3   Completion  task( 2)( 0)    task( 4)( 0)    2   0   0
3   task( 4) is running
4   task( 4) is running
5   Completion  task( 4)( 0)    task( 3)( 0)    5   2   1
5   task( 3) is running
6   Completion  task( 3)( 0)    task(63)    1   0   0
7   Preemption  task(63)    task( 1)( 1)
7   task( 1) is running
8   Completion  task( 1)( 1)    task(63)    1   0   0
9   Preemption  task(63)    task( 2)( 1)
9   task( 2) is running
10  Completion  task( 2)( 1)    task(63)    1   0   0
12  Preemption  task(63)    task( 4)( 1)
12  task( 4) is running
13  Preemption  task( 4)( 1)    task( 1)( 2)
13  task( 1) is running
14  Completion  task( 1)( 2)    task( 4)( 1)    1   0   0
14  task( 4) is running
15  task( 4) is running
16  Completion  task( 4)( 1)    task(63)    4   1   1
17  Preemption  task(63)    task( 2)( 2)
17  task( 2) is running
18  Completion  task( 2)( 2)    task( 3)( 1)    1   0   0
18  task( 3) is running
```

[Part III] FIFO Scheduler Implementation

Objective:

To implement the non-preemptive First In First Out (FIFO) scheduling for periodic tasks, and handle the miss deadline behaviors.

Problem Definition:

Implement the following task set of periodic tasks. Add necessary code to the μC/OS-II scheduler **in the kernel level** to observe how the task suffers the schedule delay.

Periodic Task Set = {τ_{ID} (ID, arrival time, execution time, period)}

Task Set 1 = {τ_1 (1, 0, 1, 4), τ_2 (2, 1, 1, 5), τ_3 (3, 2, 1, 6)}

Task Set 2 = {τ_1 (1, 0, 1, 3), τ_2 (2, 1, 2, 7), τ_3 (3, 4, 3, 12)}

Task Set 3 = {τ_1 (1, 1, 1, 6), τ_2 (2, 1, 1, 8), τ_3 (3, 5, 1, 12), τ_4 (4, 0, 3, 12)}

※If tasks arrive simultaneously, the task with the smaller TaskID will be executed first.

Evaluation:

The output format:

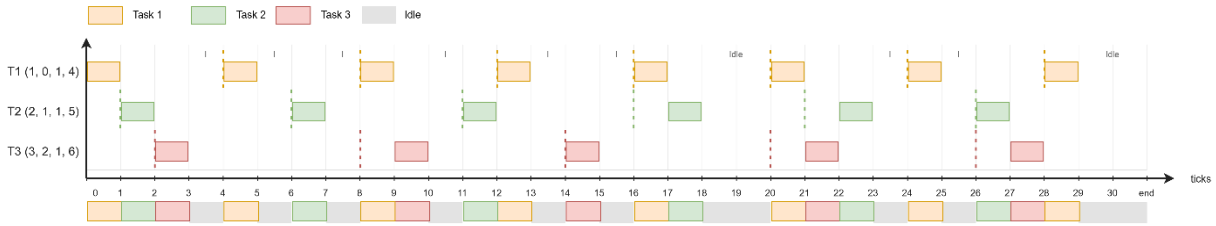
Tick	Event	CurrentTask ID	NextTask ID	Response Time	Preemption Time	Preemption Instance
##	Preemption	task(ID)(job number)	task(ID)(job number)			
##	Completion	task(ID)(job number)	task(ID)(job number)	##	##	##
##	MissDeadline	task(ID)(job number)	----------			
##	task(ID) is running					

※ If the task is Idle Task, print "*task(priority)*".

※ **You have to print and output "task(ID) is running" message in task() function.**

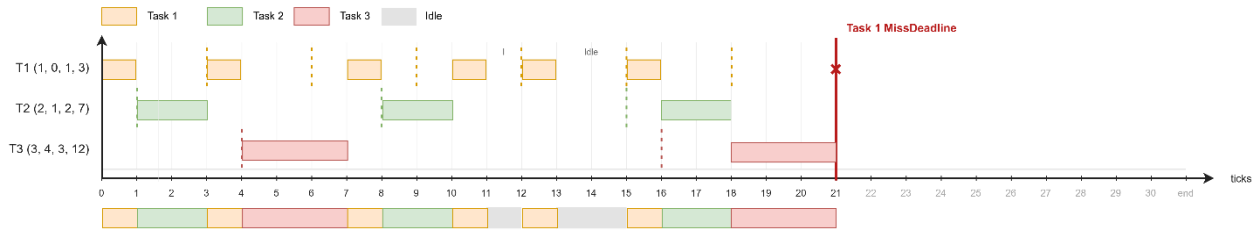
※ **Output priority: MissDeadline > Preemption > Completion > task(ID) is running .**

The output results of Task Set 1 = {τ_1 (1, 0, 1, 4), τ_2 (2, 1, 1, 5), τ_3 (3, 2, 1, 6)}:



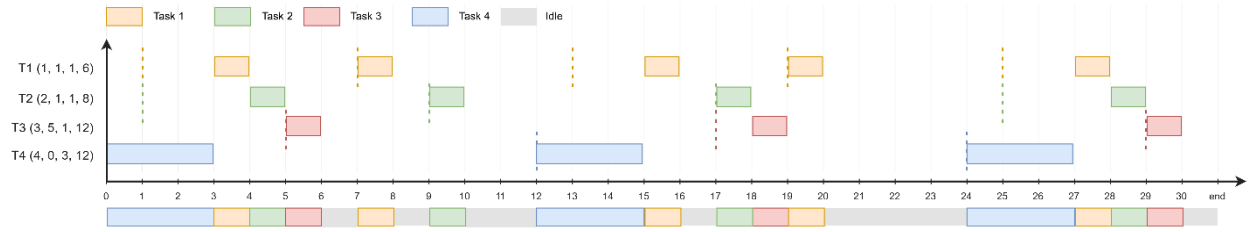

```
0   task( 1) is running
1   Completion  task( 1)( 0)    task( 2)( 0)    1   0   0
1   task( 2) is running
2   Completion  task( 2)( 0)    task( 3)( 0)    1   0   0
2   task( 3) is running
3   Completion  task( 3)( 0)    task(63)    1   0   0
4   Preemption  task(63)    task( 1)( 1)
4   task( 1) is running
5   Completion  task( 1)( 1)    task(63)    1   0   0
6   Preemption  task(63)    task( 2)( 1)
6   task( 2) is running
7   Completion  task( 2)( 1)    task(63)    1   0   0
8   Preemption  task(63)    task( 1)( 2)
8   task( 1) is running
9   Completion  task( 1)( 2)    task( 3)( 1)    1   0   0
9   task( 3) is running
10  Completion  task( 3)( 1)    task(63)    2   0   0
11  Preemption  task(63)    task( 2)( 2)
11  task( 2) is running
12  Completion  task( 2)( 2)    task( 1)( 3)    1   0   0
12  task( 1) is running
13  Completion  task( 1)( 3)    task(63)    1   0   0
14  Preemption  task(63)    task( 3)( 2)
14  task( 3) is running
15  Completion  task( 3)( 2)    task(63)    1   0   0
```

The output results of Task Set 2 = {τ_1 (1, 0, 1, 3), τ_2 (2, 1, 2, 7), τ_3 (3, 4, 3, 12)} :




```
0   task( 1) is running
1   Completion  task( 1)( 0)    task( 2)( 0)    1   0   0
1   task( 2) is running
2   task( 2) is running
3   Completion  task( 2)( 0)    task( 1)( 1)    2   0   0
3   task( 1) is running
4   Completion  task( 1)( 1)    task( 3)( 0)    1   0   0
4   task( 3) is running
5   task( 3) is running
6   task( 3) is running
7   Completion  task( 3)( 0)    task( 1)( 2)    3   0   0
7   task( 1) is running
8   Completion  task( 1)( 2)    task( 2)( 1)    2   0   0
8   task( 2) is running
9   task( 2) is running
10  Completion  task( 2)( 1)    task( 1)( 3)    2   0   0
10  task( 1) is running
11  Completion  task( 1)( 3)    task(63)    2   0   0
12  Preemption  task(63)    task( 1)( 4)
12  task( 1) is running
13  Completion  task( 1)( 4)    task(63)    1   0   0
15  Preemption  task(63)    task( 1)( 5)
15  task( 1) is running
16  Completion  task( 1)( 5)    task( 2)( 2)    1   0   0
16  task( 2) is running
17  task( 2) is running
18  Completion  task( 2)( 2)    task( 3)( 1)    3   0   0
18  task( 3) is running
19  task( 3) is running
20  task( 3) is running
21  MissDeadline    task( 1)( 6)    --------------------
```

The output results of Task Set 3 = {τ_1 (1, 1, 1, 6), τ_2 (2, 1, 1, 8), τ_3 (3, 5, 1, 12), τ_4 (4, 0, 3, 12)} :




```
0   task( 4) is running
1   task( 4) is running
2   task( 4) is running
3   Completion  task( 4)( 0)    task( 1)( 0)    3   0   0
3   task( 1) is running
4   Completion  task( 1)( 0)    task( 2)( 0)    3   0   0
4   task( 2) is running
5   Completion  task( 2)( 0)    task( 3)( 0)    4   0   0
5   task( 3) is running
6   Completion  task( 3)( 0)    task(63)    1   0   0
7   Preemption  task(63)    task( 1)( 1)
7   task( 1) is running
8   Completion  task( 1)( 1)    task(63)    1   0   0
9   Preemption  task(63)    task( 2)( 1)
9   task( 2) is running
10  Completion  task( 2)( 1)    task(63)    1   0   0
12  Preemption  task(63)    task( 4)( 1)
12  task( 4) is running
13  task( 4) is running
14  task( 4) is running
15  Completion  task( 4)( 1)    task( 1)( 2)    3   0   0
15  task( 1) is running
16  Completion  task( 1)( 2)    task(63)    3   0   0
17  Preemption  task(63)    task( 2)( 2)
17  task( 2) is running
18  Completion  task( 2)( 2)    task( 3)( 1)    1   0   0
18  task( 3) is running
19  Completion  task( 3)( 1)    task( 1)( 3)    2   0   0
```

Credit:

[PART I] Task Control Block Linked List [10%]

- The screenshot results. (5%)
- A report that describes your implementation (please attach the screenshot of the code and **MARK** the modified part). (5%)

[PART II] RM Scheduler Implementation [65%]

- The correctness of schedule results of examples (**Output.txt**). Note the testing task set might not be the same as the given example task set. (5%)
- A report that describes your implementation (please attach the screenshot of the code and **MARK** the modified part). (40%)
  - Indicate the modified parts in the given example flowchart and provide reasons for their placement within the process.
- Tests on the system will be done with the OS_TICKS_PER_SEC increased gradually. The higher value it can run while creating correct output, the higher this value will be (15%)
- Implement and describe how to handle the deadline missing situation under RM. (5%)

[PART III] FIFO Scheduler Implementation [10%]

- The correctness of schedule results of examples (**Output.txt**). Note the testing task set might not be the same as the given example task set. (5%)
- Implement FIFO and compare the schedule results with that of RM (please attach the screenshot of the code and **MARK** the modified part). (5%)

[PART IV] Discussion [15%].

- What are the parts of the kernel that has the biggest overhead? Propose methods that can be used to reduce the kernel overhead.

※ You must modify the source code!

※ Standard input and output filenames in the project are necessary for the checker. Please check the file names before submitting.

```
#define INPUT_FILE_NAME "./TaskSet.txt"
#define OUTPUT_FILE_NAME "./Output.txt"
```

※ Please set the system end time as 30 ticks in this project.

```
#define SYSTEM_END_TIME 30
```

※ You must define the algorithm in ucos_ii.h.

```
#define RM 0
#define FIFO 1
#define ALGORITHM RM/FIFO
```

※ We will use different task sets to verify your code.

※ We will check your answer on Output.txt.

Project submit:

Submit to Moodle.

Submit deadline: November 5th, 2026 (Thursday) 08:00

File name format: RTOS_Myyyddxxx_PA1.zip

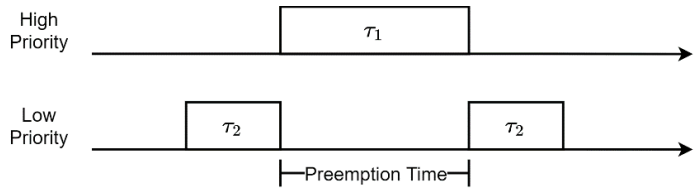
RTOS_Myyyddxxx_PA1.zip includes (The tree structure of files is shown as hints):

- The report (RTOS_Myyyddxxx_PA1.pdf).
- Folder with the executable project (RTOS_Myyyddxxx_PA1).

※ Plagiarizing is strictly prohibited.

Hints:

1. Preemption time is introduced in multiple tasking.



2. Print and write to output .txt file function

```c
#define LOG_CONSOLE 1
#define LOG_FILE    2
#define LOG_BOTH    3

void LOG_print( INT8U LOG_type, char *outfile,
                _In_z_ _Printf_format_string_ char const* const _Format,
                ...){
    va_list _ArgList;
    __crt_va_start(_ArgList, _Format);

    if((LOG_type & LOG_CONSOLE))
        vprintf(_Format, _ArgList);

    if((LOG_type & LOG_FILE) && ((Output_err = fopen_s(&Output_fp, outfile, "a")) == 0)){
        vfprintf(Output_fp, _Format, _ArgList);
        fclose(Output_fp);
    }
    return;
}
```

3. RTOS_Myyyddxxx_PA1.zip include files as follows: (PDF and the folder of the project)

